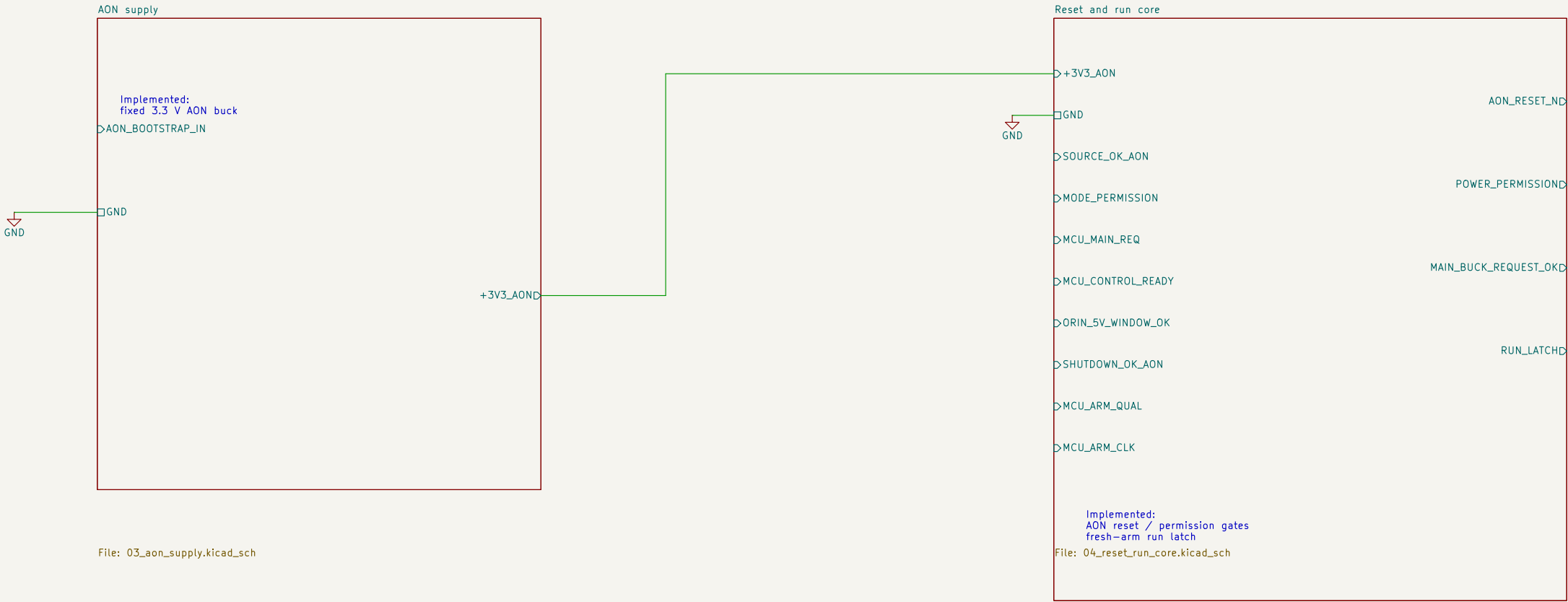


EMBER PAYLOAD COMPUTE – NEXT PROTOTYPE / STAGE 1

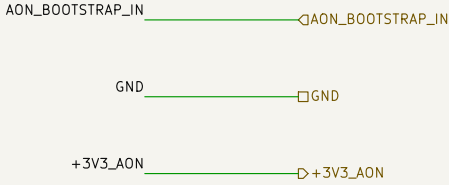
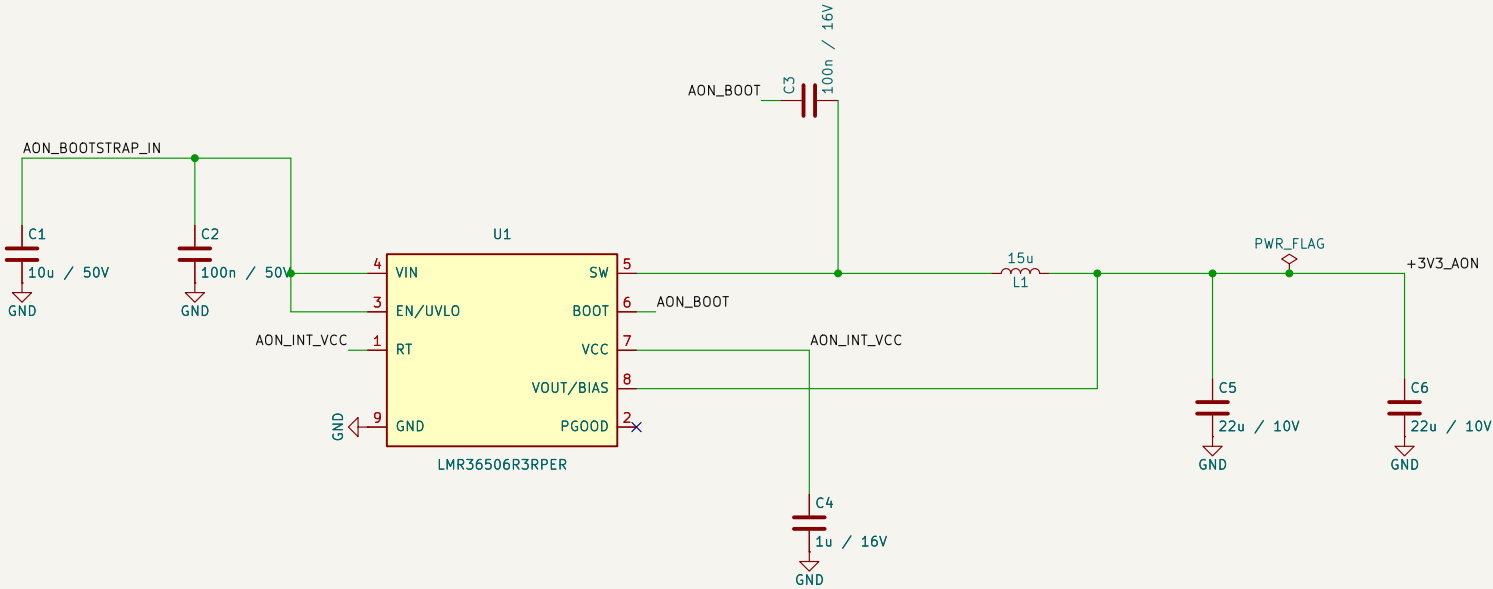


BOUNDARIES OPEN IN THIS STAGE
AON_BOOTSTRAP_IN: protected, current-limited, reverse-isolated feeds and return still required.
SOURCE_OK_AON: immutable source + raw windows. MODE_PERMISSION: sampled bench / live IHU permission.
MCU_CONTROL_READY: watchdog-qualified readiness; not a stuck-high GPIO. MCU request/arm: supervisor sheet pending.
ORIN_5V_WINDOW_OK and SHUTDOWN_OK_AON: qualified 3.3 V AON receivers; missing inputs default low.
MAIN_BUCK_REQUEST_OK: logic request ONLY; partial-power-safe buck EN driver still required.
RUN_LATCH: logic permission ONLY; Jetson POWER_EN and peripheral/reset interfaces still required.

STAGED CAPTURE: NOT LAYOUT, FAB OR FLIGHT RELEASE
No stack +3V3/+5V is joined to +3V3_AON. Main +5V_ORIN, +3V3_PERIPH and +1V8_IO are future domains.
USB lab target: 15 V / 3 A; 5 V supervision only. Conservative 7 W bring-up; 15 W current-limit qualification open.
Future sheets: connectors / PD / protected entry / source lock / regulators / STM32+CAN / Jetson / cameras / bench debug.
Existing carrier copper and disposable package/circuit studies remain untouched.

Sheet: / File: payload_compute_rev_next.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 3/3

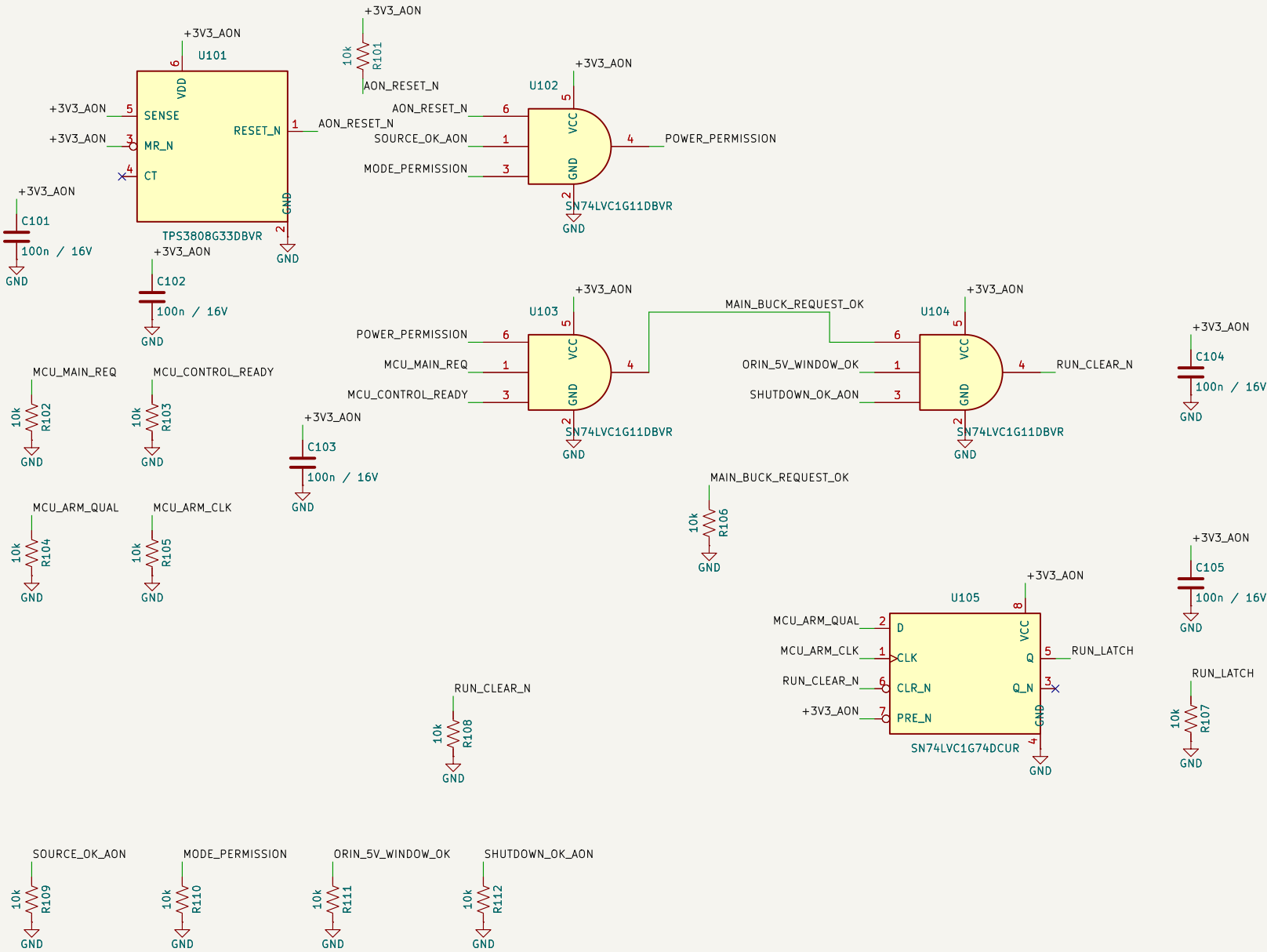
03 – ALWAYS-ON SUPPLY / IMPLEMENTED CORE



RT tied to internal VCC; nominal 1 MHz. VCC supplies only RT selection and C4; no external load.
15 uH, 2 x 22 uF output are nominal prototype values; effective C / saturation / ripple / MPN remain to qualify.
PGOOD unused. AON reset is supervised separately by TPS3808G33. EN follows protected bootstrap feed.

INTERFACE / ASSEMBLY GATES
AON_BOOTSTRAP_IN is NOT raw stack or unprotected USB. Independent limited reverse-isolated bootstrap feeds are pending.
Chosen fixed-output DC allocation: 3.2505–3.3825 V; max G33 release 3.1928 V; static margin 57.7 mV.
Startup budget target 0.200 W (old ceiling 0.250 W); transients, lower-feed startup and attach current unmeasured.
PWR_FLAG on generated +3V3_AON declares the real buck output through L1; no flag declares missing input/return.
Accepted RPE prototype lands differ from TI contour; common 0.100 / 0.125 mm stencil still to reconcile.

04 – AON RESET / PERMISSION / FRESH-ARM RUN CORE



POWER_PERMISSION = AON_RESET_N & SOURCE_OK_AON & MODE_PERMISSION
MAIN_BUCK_REQUEST_OK = POWER_PERMISSION & MCU_MAIN_REQ & MCU_CONTROL_READY
RUN_CLEAR_N = MAIN_BUCK_REQUEST_OK & ORIN_5V_WINDOW_OK & SHUTDOWN_OK_AON
RUN_LATCH samples MCU_ARM_QUAL on rising MCU_ARM_CLK only while RUN_CLEAR_N is high.
Fault/recovery does not generate an arm edge. PRE_N tied high; Q_N and TPS3808 CT intentionally NC.

BOUNDARY / TIMING CONTRACT
All external high-true qualifiers and MCU controls have 10k pulldowns (about 0.342 mA worst at 3.3825 V / 9.9k).
Future drivers must meet LVC VIH >= 2.0 V / VIL <= 0.8 V with these loads, ramps and partial power checked.
CT open: AON release 12-28 ms; not source/reference/module readiness. MR_N tied high.
Arm: keep CLK low, set D, wait >= 1 ms, pulse CLK >= 1 ms, lower CLK then D. Race/recovery tests remain.
Logic request is not physical buck EN. RUN_LATCH is not direct 5 V Jetson POWER_EN.

